

NEEBA MARY THOMAS

Data Scientist |Python Developer| ML | | Data Analyst

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TECHNICAL SKILLS

Machine Learning & Stats:

- PyTorch
- Scikit-Learn
- Pandas
- NumPy
- MLOps
- Predictive Modeling (Credit Risk / EWS)
- Statistical Analysis.
- Computer vision
- NLP

Data Engineering & Big Data:

- ETL Pipelines
- Apache Kafka
- Data Pipeline Design

Programming & Backend:

- Python (Django REST Framework, Flask, REST APIs)
- Docker
- AWS
- JavaScript
- Microservices

About Me

Data Scientist with over 2 years of experience building end-to-end ML pipelines, from data extraction and feature engineering to predictive modeling and production deployment. Highly skilled in Python, SQL, and Deep Learning frameworks (PyTorch), with a proven track record of deploying robust REST APIs and Apache Kafka event streams on AWS. Leverages a strong quantitative background (M.Sc. Applied Physics) to translate complex, messy datasets into highly accurate classification and predictive algorithms that drive business value.

WORK EXPERIENCE

NECTAR, Ernakulam, Kerala — April 2024 – March 2026

Data Scientist

- Developed RESTful backend APIs using Django REST Framework (DRF) with Mixins and Generic Views to support authentication, data processing, and CRUD operations.
- Built backend APIs for machine Run Hour calculation, including login/authentication, operational data processing, and calculation of Run Hours based on machine status data.
- Engineered automated Python (Pandas) ETL pipelines to clean, transform, validate, and standardize 100K+ unstructured operational IoT records for enterprise reporting and downstream analytics.
- Developed a real-time MQTT data monitoring tool using Python, automating diagnostic workflows and reducing manual data-processing effort by 60%.
- Hands-on experience with IoT data processing and MQTT-based real-time communication, including monitoring machine/operational data streams, processing JSON payloads, and integrating IoT data into Python-based analytics and backend workflows.
- Analyzed and cleaned 100K+ operational IoT records** and developed **Random Forest Regression–based anomaly detection workflows** to identify abnormal equipment behavior and support fault diagnostics, including **Low Delta T logic**, improving fault identification by 25%.
- Engineered a Docker-containerized, microservice-driven Work Order system using React, Django REST Framework, and PostgreSQL, featuring custom ORM analytics APIs to track real-time technician performance metrics across a scalable Linux environment.

Python Full-Stack Development Intern

Inmakes Infotech Pvt. Ltd., Kerala — Aug 2023 – Nov 2023

- Built end-to-end web applications with Python (Django, Flask) and REST APIs, applying Object-Oriented Programming (OOP) principles and SQL-driven data analysis.

Databases & Data Visualization :

- Advanced SQL
- PostgreSQL
- Power BI(Power Query, DAX, RLS)
- Tableau
- Excel
- Matplotlib
- Seaborn

EDUCATION

M.Sc. Applied Physics

M.G. University, Kottayam
CMS College 2015 - 2017

B.Sc. Physics

M.G. University, Kottayam
KG College 2012 - 2015

PROFILES

LinkedIn

GitHub

LANGUAGES

English

Hindi

Malayalam

Language Trainer (IELTS) 2020 – 2024

Manjoorans, Langtrees, Trinity, IRS Group

- Utilized data-driven performance tracking to optimize coaching while concurrently upskilling in Python, SQL, and Machine Learning.

MGM College, Puthuppally 2017 – 2019

Science Teacher

- Taught core sciences, leveraging a background in physics to cultivate strong analytical and mathematical problem-solving skills

PROJECTS

Credit Risk Early Warning System (EWS)

PyTorch, Kafka, AWS, Django, Scikit-Learn, React

- Engineered a predictive Early Warning System using PyTorch and Scikit-Learn, featuring a robust preprocessing "firewall" to dynamically sanitize applicant data and flag high-risk loan defaults.
- Architected a dual-pathway AWS pipeline, combining a synchronous Django REST API for real-time risk scoring with an asynchronous Apache Kafka (KRaft) queue for high-volume batch processing.
- Built a dynamic React frontend equipped with an automated PostgreSQL polling mechanism to track background worker progress and deliver real-time AI processing metrics to end users.

Automated ETL Pipeline & Data Warehouse

Python, PostgreSQL, REST APIs

- Built an automated Python ETL pipeline to extract JSON data from a public REST API, transform it with Pandas, and load it into a PostgreSQL Star Schema with referential integrity.
- Implemented rigorous source-to-target reconciliation checks (row counts, null checks) and an SQL audit framework to proactively catch and investigate data anomalies.

Industrial ETL Pipeline & EWS Dashboard

Python, ,PostgreSQL, Power BI, Power Query, DAX, RLS

Built a telemetry data pipeline with Power BI dashboarding and Row-Level Security for role-based access.

Customer Churn ML Pipeline

Python, Scikit-Learn,

Built a churn-prediction pipeline using SMOTE for class imbalance and K-Means clustering for customer segmentation